

Software Engineering Internship & Placement Roadmap

Phase 1: Programming Fundamentals (Week 1)

1. ~~Number Problems (Easy)~~

These build your logical thinking.

Topics:

- Count Digits 
- Reverse Number
- Sum of Digits
- Palindrome Number
- Armstrong Number
- Prime Number
- Perfect Number
- Strong Number
- GCD & LCM
- Factorial
- Fibonacci
- Count Even/Odd Digits
- Decimal $\leftrightarrow$ Binary Conversion

Interview Questions

- Check whether a number is prime.
- Find GCD using Euclid Algorithm.
- Print first N Fibonacci numbers.
- Reverse an integer without converting to String.

2. ~~Pattern Printing (Easy)~~

Companies rarely ask patterns directly, but they improve loop understanding.

Topics

- Square
- Rectangle
- Right Triangle
- Inverted Triangle
- Pyramid
- Diamond
- Hollow Square
- Hollow Pyramid
- Floyd Triangle
- Pascal Triangle

Focus

- Nested loops
- Space handling
- Mathematical patterns

~~Phase 2: Arrays (Very Important~~ ★ ★ ★ ★ ★)

Arrays appear in almost every coding interview.

Topics

Basics

- Input
- Output
- Traversal

Searching

- Maximum

- [Minimum](#)
- [Second Largest](#)
- [Second Smallest](#)

Rearrangement

- [Reverse Array](#)
- [Left Rotation](#)
- [Right Rotation](#)
- [Rotate by K](#)
- [Move Zeroes](#)

Frequency

- [Count Frequency](#)
- [Remove Duplicates](#)

Merge

- [Merge Arrays](#)
- [Merge Sorted Arrays](#)

Important Problems

- [Missing Number](#)
- [Leaders in Array](#)
- [Prefix Sum](#)
- [Kadane Algorithm](#)
- [Equilibrium Index](#)
- [Stock Buy & Sell](#)
- [Majority Element](#)
- [Trapping Rain Water](#)
- [Product Except Self](#)

Interview Level

Easy → Medium

Phase 3: Strings (Very Important ★★★★★)

Topics

Basics

- String Input
- Character Traversal

Counting

- Vowels
- Consonants
- Digits
- Words

Manipulation

- Reverse String
- Toggle Case
- Remove Spaces
- Remove Extra Spaces
- Replace Character

Algorithms

- Palindrome
- Anagram
- Character Frequency
- Longest Word
- String Compression
- Longest Common Prefix
- Longest Palindromic Substring

Interview Questions

- Reverse words in sentence.
 - Check Anagram.
 - Compress String.
 - Count occurrences.
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~~Phase 4: Matrix (Medium)~~

Topics

- Matrix Input
- Matrix Output
- Addition
- Multiplication
- Transpose
- Diagonal Sum
- Row Sum
- Column Sum
- Identity Matrix
- Symmetric Matrix
- Rotate Matrix

Interview Questions

- Rotate matrix by 90°
 - Spiral Traversal  
 - Search in Sorted Matrix
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Phase 5: Functions

Topics

- Pass by Value
- Pass by Reference
- Function Overloading
- Default Parameters
- Recursive Functions

Interview

Difference between

- Stack Memory
- Heap Memory
- Function Call Stack

Phase 6: Recursion ★★★★★

Topics

- Print 1→N
- Print N→1
- Sum
- Factorial
- Fibonacci
- Reverse Array
- Palindrome
- Binary Search
- Tower of Hanoi

Advanced

- Backtracking Introduction
- Generate Subsets
- Permutations

~~Phase 7: Sorting~~ ★ ★ ★ ★ ★

Know

Bubble Sort

- Time Complexity
- Stable
- In-place

Selection Sort

Insertion Sort

Merge Sort

Quick Sort

Counting Sort

Extra

- Heap Sort
- Radix Sort

Interview

Compare every sorting algorithm.

~~Phase 8: Searching~~

Topics

Linear Search

Binary Search

Lower Bound

Upper Bound

Binary Search on Answer

Examples

- Square Root
 - Allocate Books
 - Aggressive Cows
 - Capacity to Ship Packages
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Phase 9: STL (C++) / Collections (Java)

Vector / ArrayList

Know

- push
- pop
- insert
- erase

Map

- Ordered Map
- Unordered Map

Set

- Set
- Multiset

Queue

Stack

Priority Queue

Deque

Algorithms

sort()

reverse()

find()

count()
lower_bound()
upper_bound()
max_element()
min_element()
binary_search()

Phase 10: Mathematics

~~Topics~~

~~GCD~~

~~LCM~~

~~Prime Numbers~~

Sieve

Fast Power

Modular Arithmetic

Modulo Properties

Combination

Permutation

~~Phase 11: Bit Manipulation~~ ★★★★★

Topics

Even Odd

Set Bit

Clear Bit

Toggle Bit

Count Bits

Power of Two

XOR

Right Shift

Left Shift

Problems

Single Number

Missing Number

Subset Generation

~~Phase 12: Hashing~~ ★ ★ ★ ★ ★

Topics

Frequency Array

HashMap

HashSet

Applications

Duplicate Detection

Count Frequency

Two Sum

Longest Consecutive Sequence

1. Frequency Count
2. Contains Duplicate
3. Two Sum
4. Majority Element
5. Valid Anagram
6. First Unique Character
7. Group Anagrams
8. Top K Frequent Elements
9. Longest Consecutive Sequence
10. Subarray Sum Equals K
11. Longest Subarray with Sum K
12. Minimum Window Substring

~~Phase 13: Two Pointers~~

Topics

Two Sum

Pair Sum

Remove Duplicates

Container With Most Water

Three Sum

~~Phase 14: Sliding Window~~ ★ ★ ★ ★ ★

Topics

Maximum Sum Subarray

Longest Substring

Minimum Window

Fixed Window

Variable Window

Interview Favorite

Phase 15: Greedy Algorithms

Topics

~~Activity Selection~~

~~Job Sequencing~~

Fractional Knapsack

Minimum Platforms

Meeting Rooms

~~Phase 16: Binary Search on Answer~~

Topics

Allocate Books

Painter Partition

Aggressive Cows

Koko Eating Bananas

Capacity to Ship Packages

Phase 17: Dynamic Programming (DP)

Beginner

Fibonacci

Climbing Stairs

House Robber

Coin Change

Knapsack

Longest Common Subsequence

Longest Increasing Subsequence

Matrix Chain Multiplication

~~Phase 18: Linked List~~ ★ ★ ★ ★ ★

~~Topics~~

~~Singly Linked List~~

~~Doubly Linked List~~

~~Circular Linked List~~

~~Reverse List~~

Middle Node

~~Detect Cycle~~

Merge Lists

Remove Nth Node

~~Phase 19: Stack~~

Problems

Balanced Parentheses

Next Greater Element

Min Stack

Infix to Postfix

Postfix Evaluation

Largest Rectangle in Histogram

Phase 20: Queue

Topics

Queue

Circular Queue

Deque

Priority Queue

Applications

BFS

Task Scheduling

Phase 21: Trees ★ ★ ★ ★ ★

Topics

Binary Tree

Binary Search Tree

Traversal

- Preorder
- Inorder
- Postorder
- Level Order

Problems

Height

Diameter

Lowest Common Ancestor

Balanced Tree

Maximum Path Sum

Phase 22: Heap

Topics

Min Heap

Max Heap

Priority Queue

Problems

Top K Elements

Kth Largest

Merge K Lists

Problem	Heap
Top K Largest	✓ Min Heap (size = k)
Top K Smallest	✓ Max Heap (size = k)
Kth Largest	✓ Min Heap (size = k)
Kth Smallest	✓ Max Heap (size = k)

~~Phase 23: Graph~~ ★★★★★

Topics

Representation

BFS

DFS

Connected Components

Cycle Detection

Topological Sort

Shortest Path

- Dijkstra

- Bellman Ford
- Floyd Warshall

Minimum Spanning Tree

- Kruskal
- Prim

Disjoint Set Union (Union-Find)

Phase 24: Object-Oriented Programming (Especially for Java)

Topics

- Class & Object
- Constructor
- Method Overloading
- Method Overriding
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction
- Interface
- Abstract Class
- this and super
- Static Members
- Exception Handling
- Collections Framework
- Generics
- File I/O
- Multithreading (Basics)

Interview Focus:

- Explain the four pillars of OOP with examples.
 - Difference between interface and abstract class.
 - Difference between overloading and overriding.
 - HashMap vs Hashtable, ArrayList vs LinkedList.
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Phase 25: SQL & Database ★★★★★

Topics

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- JOIN (INNER, LEFT, RIGHT, FULL)
- SELF JOIN
- Aggregate Functions
- Subqueries
- CTE
- Window Functions (ROW_NUMBER, RANK, DENSE_RANK)
- Normalization
- Indexes
- Transactions
- ACID Properties

Common Interview Problems:

- Second highest salary
- Nth highest salary

- Department-wise highest salary
 - Duplicate records
 - Employees without managers
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Phase 26: Operating System Basics

Topics

- Process vs Thread
 - CPU Scheduling
 - Deadlock
 - Mutex & Semaphore
 - Virtual Memory
 - Paging
 - Segmentation
 - Context Switching
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Phase 27: Computer Networks

Topics

- OSI Model
 - TCP/IP
 - HTTP vs HTTPS
 - DNS
 - IP Addressing
 - Cookies & Sessions
 - REST API Basics
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Phase 28: Software Engineering & Git

Topics

- SDLC
 - Agile & Scrum
 - Git Commands (clone, add, commit, push, pull, branch, merge)
 - GitHub Workflow
 - Basic Design Patterns (Singleton, Factory, MVC)
 - Unit Testing Basics
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Phase 29: Resume & Projects

A strong internship profile should include:

- 2–3 well-documented projects (e.g., your Dishari Medical Portal, Staff Management System, Mario Game).
 - Clean GitHub repositories with README files.
 - A one-page ATS-friendly resume.
 - Problem-solving practice (150–300 coding questions).
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Phase 30: Mock Interviews & Practice

Practice on:

- LeetCode (Easy → Medium)
- GeeksforGeeks
- HackerRank
- Codeforces (beginner contests)

Target milestones:

- **100 Easy** problems
- **150 Medium** problems
- **20–30 Hard** problems (optional for most internships)

Recommended Study Order

1. Programming Basics & Number Problems
2. Pattern Printing
3. Arrays
4. Strings
5. Functions & Recursion
6. Sorting & Searching
7. STL (C++) / Java Collections
8. Mathematics & Bit Manipulation
9. Hashing
10. Two Pointers & Sliding Window
11. Linked Lists
12. Stack & Queue
13. Trees
14. Heap
15. Graphs
16. Dynamic Programming
17. OOP
18. SQL
19. Operating Systems
20. Computer Networks
21. Software Engineering & Git
22. Resume, Projects, and Mock Interviews

If you complete this roadmap, you'll have covered the core topics commonly assessed in internship and entry-level software engineering interviews at product companies, startups, and service-based companies.

